

Dr. Shadia Yahya Baroud

AI Researcher & Assistant Professor

*Multi-Agent Systems · Federated Learning · Agentic AI · Privacy-Preserving
AI · Reinforcement Learning · Distributed Intelligent Systems*

IEEE Senior Member | ITU AI for Good Scholar

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RESEARCH & PROFESSIONAL HIGHLIGHTS

PhD in Computing (AI) UNIMY, Malaysia · 2025	16+ Publications <i>Q1</i> Journals · IEEE · CRC Press	Active Grants Pipeline	IEEE Senior Member Since 2025
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Professional Summary

AI researcher with a PhD in Computing (Artificial Intelligence) and over 15 years of multidisciplinary experience designing intelligent, privacy-preserving and resource-aware systems that bridge theory and real-world deployment – across federated learning, multi-agent coordination and generative AI under uncertainty. Background spans nine years as a database developer and systems analyst before transitioning fully into AI research, giving my technical framework design a systems-engineering grounding rather than a purely theoretical one.

Research output includes 16 peer-reviewed outputs spanning Q1 journals, IEEE proceedings and a CRC Press book chapter. Verified citation profile (Google Scholar): **82 total citations** | **h-index: 4** | **i10-index: 2**, with growth from 4 (2023) to 34 (2024) to 80 (2025/26). Active international collaborations with the University of Glasgow (EPSRC TransiT) reflect a sustained commitment to high-impact, globally engaged scholarship.

IEEE Senior Member and two-time ITU AI for Good Scholar (Geneva, 2024 & 2025). Currently Assistant Professor at Al-Aqsa University, delivering AI, Machine Learning, Business Intelligence, Research Methodology & Academic Writing and Software Engineering to 600+ undergraduate and postgraduate students and supervising over 200 Final Year Projects and postgraduate students to date.

Research focus: Privacy-preserving federated learning · Multi-agent coordination for distributed AI · Generative models under uncertainty · AI for healthcare and safety-critical systems · Smart manufacturing & intelligent infrastructure.

Academic & Research Experience

Senior Lecturer & Assistant Professor – AI & Computer Science | Al-Aqsa University, Palestine | 02/01/2025 – Present (Part-time)

- Delivering AI, Machine Learning, Robotics, Business Intelligence and Software Engineering courses to 600+ undergraduate and postgraduate students.
- Supervising 100+ Final Year Projects (FYP) annually, guiding students through research design, implementation and technical presentation – several achieving publication-quality outcomes.
- Mentoring postgraduate Master's students in research methodology, academic writing and thesis development.
- Developing and updating course syllabi aligned with Industry 4.0 trends, emerging AI technologies and institutional accreditation standards.
- Curriculum development leadership.** Member of three university-level committees developing new degree programmes: MSc in Artificial Intelligence (chaired by Dr. Yousef Hammouda, curriculum submitted to the Palestinian Ministry of Higher Education), BSc in Artificial Intelligence (chaired by Dr. Mohammed Awadallah) and BSc in Web Design and Development (chaired by Dr. Abdelrafe Elzamy).
- Departmental research seminars (2025/26).** Delivered sessions on research methodology, academic writing and the foundations of multi-agent systems, linked to the ML2MAS framework.

Research Assistant – B5G URLLC & Wireless Communications | Sunway University, Malaysia | 02/01/2023 – 02/01/2024 (*Contract*)

- Contributed to an industry/government co-funded FRGS/IRGS grant on ultra-reliable low-latency communication (URLLC) for Beyond 5G (B5G) smart manufacturing networks, under Dr. Athirah Mohd Ramly.
- Contributed to SCMA-OFDM physical layer system design incorporating polar FEC codes, targeting 99.99999% packet success rate and $\leq 1\text{ms}$ end-to-end latency per B5G/3GPP standards.
- Conducted channel estimation analysis using LS and MMSE techniques under Rayleigh fading channel models; literature reviews on NOMA, grant-free access and URLLC for IIoT environments.

Remained continuously research-active throughout 2024: completed PhD Research modules 7–8 toward thesis submission under Prof. Dr. Nor Adnan Yahaya, presented at four IEEE/ITU venues, and served as an ITU AI for Good Scholar (Geneva, 2024 cycle).

Research Fellow – AI & Data Science | UNIMY (University Malaysia of Computer Science & Engineering), Petaling Jaya | 30/04/2021 – 01/11/2022 (*Contract*)

- Conducted PhD-level research in multi-agent systems, federated learning and applied AI under Prof. Dr. Nor Adnan Yahaya; lead/first author on the majority of resulting publications.
- Developed the ML2MAS framework (Q1, Sustainable Operations and Computers, 2025): a multi-agent coordination architecture integrating Bayesian Neural Networks with GANs for uncertainty-aware decision support in smart manufacturing.
- Co-developed EcoTransTwin – a federated learning framework for distributed, privacy-preserving model training across heterogeneous edge nodes in sustainable transport – with Dr. Paul Harvey (University of Glasgow).
- Accessed HPC infrastructure at UNIMY and Sunway University for large-scale AI experiments and distributed systems research.
- Secured over £5,000 in competitive international conference and scholar funding (ITU Geneva, WAIC Shanghai, IEEE).
- Supported Prof. Dr. Yahaya's supervision of undergraduate and Master's students, contributing to research design, methodology and academic writing.
- Built an active international research network across Malaysia, Palestine, the UK (Glasgow) and Switzerland (Geneva).

Database Developer & Systems Analyst | Gaza, Palestine | 01/07/2019 – 29/04/2021 (*Full-time*)

- Designed relational database systems supporting large-scale administrative and operational workflows; conducted system analysis and requirements gathering; developed SQL queries, stored procedures and reporting pipelines.

IT Instructor / Trainer | Khan Younis Training College (KYTC) – UNRWA / United Nations, Gaza | 18/10/2017 – 30/06/2019 (*Contract*)

- Delivered structured IT training aligned with UN educational standards; designed instructional materials supporting employability and digital literacy; assessed learner progress.

Database Developer & Systems Analyst | Diwan, Gaza, Palestine | 27/02/2012 – 17/10/2017 (*Full-time*)

- Designed and developed relational database systems supporting large-scale administrative and operational workflows; maintained data integrity, system security and cross-project documentation.

Lecturer – AI & Computer Science | Al-Aqsa University, Palestine | 18/10/2010 – 30/06/2016 (*Part-time*)

- Taught Web Design, E-Commerce, Multimedia and Software Engineering across undergraduate programmes.

Independent Researcher, AI Consultant & Educator | Self-employed, Gaza, Palestine/Malaysia | 2010 – Present (*Freelance, ongoing alongside above roles*)

- Applied AI and systems research; many published papers originate from or are directly informed by real-world problems encountered through this practice.

- Research consultancy on smart contract formal verification using Solidity-based verifiers.
- AI framework design and consulting services to clients across the GCC and MENA region.
- Design and delivery of online training programmes in AI, programming, data science and IT fundamentals to learners across the world.

Education

Doctor of Philosophy (PhD) – Computing (Artificial Intelligence) | UNIMY — University Malaysia of Computer Science and Engineering, Petaling Jaya, Malaysia | *Commenced 05/2021 | Awarded 19/08/2025*

Thesis: "A Multi-Agent-Based Framework for Scalable Machine Learning Integration and Adaptive Decision Support in Smart Applications." Supervisor: Prof. Dr. Nor Adnan Yahaya. Degree conferred following Senate endorsement, Special Senate Meeting 19/08/2025.

Master of Science – Information Technology | The Islamic University of Gaza, Palestine | *Commenced 2013 | Awarded 08/05/2018*

Thesis: "Applying Case-Based Reasoning on Breast Cancer Data: Gaza Strip Case Study" — intelligent systems and data-driven clinical decision support; foundational to current healthcare AI publications. Supervisor: Prof. Alaa M. El-Halees.

Postgraduate Diploma in Education | Al-Aqsa University, Palestine | *2011 – 2012*

Professional qualification in tertiary teaching theory and pedagogy — supports curriculum design, student assessment and academic mentoring practice.

BSc – Computer Science | Al-Aqsa University, Palestine | *Commenced 2007 | Awarded 03/08/2010*

Cumulative average 84.1% (Very Good).

Diploma in Computer Technology | Gaza Training College (GTC — UNRWA/United Nations) | *2005 | Awarded 01/08/2007*

General average 84.7%.

Research & Publications

Verified citation profile (Google Scholar): 82 total citations | h-index: 4 | i10-index: 2 | 16 indexed outputs | Growth: 4 (2023) → 34 (2024) → 82 (2025/26)

Published Journal Articles

S. Y. Baroud, N. A. Yahaya and A. M. Elzamly, "Cutting-edge AI approaches with MAS for predictive maintenance in Industry 4.0: challenges and future directions," *Journal of Applied Data Sciences*, vol. 5, no. 2, pp. 455–473, 2024. [42 citations – highest-cited output]

S. Y. Baroud and N. A. Yahaya, "ML2MAS: A multi-agent reinforcement learning and BNNs-GAN integration framework for smart manufacturing optimization," *Sustainable Operations and Computers*, vol. 6, pp. 217–228, 2025. [Q1; 12 citations]

Context: ML2MAS integrates Bayesian Neural Networks with GAN-based generative modelling for uncertainty-aware decision support in smart manufacturing; provides the multi-agent coordination foundation my current research extends toward federated learning. My role: lead author - conceived the framework, designed the BNN-GAN integration, ran all experiments, led manuscript preparation.

Manuscripts Under Review

S. Y. Baroud and P. Harvey, "EcoTransTwin: An AI-based federated digital twin framework for sustainable transport decarbonization," under review, *Transportation Research Part C (Q1, Elsevier)*, 2026.

Context: federated learning framework for privacy-preserving distributed model training across heterogeneous edge nodes in sustainable transport, with the University of Glasgow's EPSRC TransiT programme. My role: lead author – designed the federated architecture and led writing; Dr. Paul Harvey contributed transport digital-twin context and review.

S. Y. Baroud, "MASHA: A multi-agent system for healthcare sentiment analysis – migraine detection in Arabic tweets," under review, *PLOS ONE*, 2026. Preprint: medRxiv, DOI: 10.64898/2026.05.21.26352626.

Context: multi-agent architecture for Arabic-language healthcare sentiment analysis, addressing an under-served research need. My role: sole author.

S. Y. Baroud, A. M. Elzamly and H. Abughali, "A decentralized multi-agent framework integrating interpretable machine learning for automated warranty fraud detection in supply chains," under review, Human-Centric Intelligent Systems (Q1, Springer Nature), 2026. Preprint: Research Square, DOI: 10.21203/rs.3.rs-10569728/v1

Context: interpretable multi-agent framework combining machine learning and human-in-the-loop review for warranty fraud detection. My role: lead author — developed the multi-agent framework and interpretability layer.

A. M. Elzamly, **S. Y. Baroud**, M. Alaqqad and S. S. Abu-Naser, "An Explainable XGBoost-SHAP Framework for Lightweight Intrusion Detection in IoT Networks," in preparation for submission, Digital Communications and Networks (Q1, KeAi/Elsevier), 2026.

Other Indexed Journals

Y. Khamayseh, N. A. Yahaya and **S. Y. Baroud**, "Comparing machine learning algorithms for intrusion detection using the UNSW-NB15 dataset," Journal of Innovation and Emerging Digital Technologies, vol. 2, no. 1, pp. 21–35, 2023.

M. N. Qureshi, N. A. Lazam, **S. Baroud** and A. M. A. Barhoom, "Optimisation and cross-validation of hybrid pricing algorithm using SL's CART model for e-hailing," Journal of Advanced Research in Applied Sciences and Engineering Technology, 2025.

S. Y. Baroud, "Consumers' attitude and purchasing behaviour on mobile shopping: A review," Journal of Innovation and Emerging Digital Technologies, vol. 1, no. 1, pp. 20–27, 2022.

N. A. Yahaya, M. Abdulgabber and **S. Y. Baroud**, "An evolutionary approach to software testing ontology development," International Journal of Information Technology, vol. 8, no. 4, pp. 1–8, 2022.

Published Conference Papers

A. Baraka and **S. Y. Baroud**, "AidVision: Dual-temporal satellite damage assessment for rapid humanitarian prioritization," HDTC'26, University of Bahrain, 2026. [Accepted]

Context: dual-temporal satellite imagery for post-disaster damage assessment and humanitarian prioritisation. My role: supervisor and co-author – supervised Ahmed Baraka (first author) through concept development, methodology and manuscript.

S. Y. Baroud, "AI-powered multi-agent system for Type 2 Diabetes care: IoT, federated learning and mobile health integration," IEEE ICMCTC, Malaysia, 2025. [Best Presenter Award]

S. Y. Baroud, A. M. El-Halees, R. S. Baraka and N. A. Yahaya, "AI-driven CBR and ontological integration framework for enhancing breast cancer care: case study," IEEE RAAI, Singapore, 2024, pp. 284–289.

S. Y. Baroud and N. A. Yahaya, "RAPID: Integrating AI and multi-agent systems for enhanced traffic management with YOLOv9," 3ICT, Bahrain (online), 2024, pp. 246–253.

S. Y. Baroud and M. N. Qureshi, "Harnessing Industry 4.0: A framework for integrating AI and emerging technologies," IEEE WAIE, Tokyo, Japan, 2024, pp. 202–209. [Best Presenter Award]

S. Y. Baroud and N. A. Yahaya, "Optimizing smart manufacturing through multi-agent integration of ML models," IEEE RIVF, Da Nang, Vietnam, 2024, pp. 341–346.

S. Y. Baroud, N. A. Yahaya and A. M. Elzamly, "Spare parts management analytics for smart maintenance," Proc. 1st ICDiTS, Malaysia, 2023. [Best Presenter Award]

Book Chapter

S. Y. Baroud, "Fault diagnosis and predictive maintenance using AI in the mechatronics industry," in Handbook of AI-Based Mechatronics Systems and Smart Solutions in Industrial Automation, CRC Press, 2025, pp. 1–18.

Policy and Technical Reports

S. Y. Baroud, contributing author, AI for Good Innovate for Impact Report 2024 & 2025, International Telecommunication Union (ITU/UN), Geneva. ISBN: 978-92-61-40961-6 (2025).

Context: use case documentation across multiple sectors of the AI for Good taxonomy, evaluated by ITU's Technical Advisory Committee; included in ITU's global AI Playbook (2026, 427 curated use cases). My role: author & reviewer - contributed cross-domain use case documentation, methodology and analysis.

Invited Presentations & International Engagement

- **Speaker – "Innovate for Impact: The Winning AI Use Cases," ITU AI for Good Global Summit 2024, Geneva.** In-person session, 30/05/2024. Panel presenting winning AI use cases, alongside international co-speakers including Prof. Ir. Dr. Aduwati Sali (Director, INSPEM, Universiti Putra Malaysia), Prof. Asrat Mulatu Beyene (Addis Ababa Science & Technology University) and Shan Xu (BRICS AI Centre); moderated by Vishnu Ram OV (ITU).
- **Panel contributor – WAIC (World Artificial Intelligence Conference) 2024, Shanghai.** Panel discussion on reviewed AI for Good use cases, as part of ITU AI for Good Scholar engagement.
- **Invited guest lecturer – IEEE Women in Engineering (WIE), Iraq Chapter, Invitation accepted; scheduled for 10/09/2026.** "Federated Learning for Multi-Agent Systems: Building Privacy-Preserving AI", based on the ML2MAS framework. Host: Dr. Ruaa Al-Falluji, Al-Nahrain University.
- **Invited trainer – International Modern Arabic School (IMAS), UNITEN, Malaysia, 18/08/2026.** Full-day professional development workshop, "AI-Driven Differentiation & Resource Generation," delivered in Arabic to approximately 25 teachers.
- **AI for Good Global Summit – Geneva, 2024 and 2025.** Participated in the UN's flagship annual summit on AI for the SDGs.
- **IEEE WCNC 2026 – KL, April 2026.** Attendee, technical volunteer, networking with the international wireless communications research community.

Recognition & Awards

- **AI for Good Scholar (2024 & 2025)** – International Telecommunication Union (ITU/UN), Geneva. Formally engaged both years under the Innovate for Impact initiative; in 2025, named among a small group of scholars recognised for outstanding efforts (from 171 applications across 37 countries, 16 scholars selected globally).
- **IEEE Senior Member (2025 – Present)** – awarded to IEEE members with substantial professional and research contributions; held by approximately 10% of IEEE membership.
- **Best Presenter Awards** – IEEE ICMCTC (2025, Malaysia), IEEE WAIE (2024, Tokyo), ICDiTS (2023, Malaysia).

Grants, Funding & Research Infrastructure

- **ITU AI for Good Summit Scholar Awards** – Geneva, Switzerland, 2024 & 2025 (approx. £3,000 total, competitive travel scholarship).
- **WAIC Shanghai Travel Scholarship** – World Artificial Intelligence Conference, 2024 (£2,000).
- **IEEE Student Member Conference Travel Grant** – 2023–24.
- **FRGS/IRGS Grant** – co-researcher, industry/government co-funded Beyond-5G URLLC research project (Sunway University, 2023).
- **HPC Access** – institutional High-Performance Computing infrastructure at UNIMY and Sunway University (2021–2025).

Supervision & Mentoring

- Supervised undergraduate final-year projects in AI, machine learning, software engineering and information systems during my appointments at Al-Aqsa University.
- Mentored postgraduate researchers in research methodology, academic writing and thesis development.
- Supported undergraduate and Master's research supervision at UNIMY under Prof. Dr. Nor Adnan Yahaya, contributing to research design, methodology and manuscript development.
- Supervised Ahmed Baraka's AidVision project through concept development, methodology and manuscript preparation, resulting in a co-authored paper accepted at HDTC'26, University of Bahrain.

Conference Organisation & Academic Service

- International Advisory Committee Member & Conference Moderator – WCASET-2025, Kuala Lumpur, November 2025.
- Programme Committee Member – IEEE and Scopus-indexed international conferences in AI and Computer Science.
- Peer Reviewer – IEEE Transactions, Scopus-indexed journals and conferences in AI, Machine Learning and Intelligent Systems (2019 – Present).
- IEEE Conference Volunteer – IEEE WCNC 2026, IEEE GLOBECOM 2023, IEEE ICIP 2023.
- ITU AI for Good Scholar Network (2024–2025) – AI safety and ethics reviews across healthcare, education, industry and agriculture.

Professional Certifications & Skills

- IEEE Senior Member – Institute of Electrical and Electronics Engineers (2025 – Present)
- ITU AI for Good Scholar – International Telecommunication Union, Geneva (2024–2025)
- CCNA – Cisco Certified Network Associate
- Professional Certificate in Data Science & AI (R & Python)
- Postgraduate Diploma in Education – Al-Aqsa University (2011–2012)
- ICDL – International Computer Driving Licence

Technical: Python · R · PyTorch · TensorFlow · Federated learning frameworks (Flower) · Multi-agent RL (MAPPO, QMIX, IPPO) · SQL / relational databases · HPC / SLURM · Git · LaTeX

Key Research Collaborators

- **Prof. Dr. Nor Adnan Yahaya** – UNIMY, PhD supervisor and co-author on 7+ publications
- **Dr. Paul Harvey** – University of Glasgow, EPSRC TransiT programme; co-author on EcoTransTwin
- **Dr. Abdelrafe Elzamly** – Al-Aqsa University, Head of Department; co-author on multiple papers
- **Dr. Athirah Mohd Ramly** – Sunway University; supervisor on B5G URLLC research
- **Prof. Alaa M. El-Halees** – Islamic University of Gaza; MSc supervisor, co-author

PROFESSIONAL REFERENCES

Prof. Dr. Nor Adnan Yahaya Professor, UNIMY / Malaysia University of Science and Technology, Malaysia, adnan.yahaya@unimy.edu.my	Assoc. Prof. Dr. Abdelrafe M. Elzamly Head of Dept., Computer & Information Sciences, Al-Aqsa University, Palestine, abd_elzamly@alagsa.edu.ps
Prof. Omer Rana Professor of Performance Engineering School of Computer Science and Informatics, Cardiff University, UK, Cross-Council Director, EPSRC National Edge AI Hub, RanaOF@cardiff.ac.uk	Dr. Paul Harvey Co-Investigator, EPSRC TransiT (£25M), University of Glasgow, UK, Paul.Harvey@glasgow.ac.uk